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Governance of forests and governance of forest information: Interlinkages in the age of open and digital data

Salla Rantala[a,*], Brent Swallow[b], Riikka Paloniemi[a], Elina Raitanen[a,1]

[a] Environmental Policy Centre, Finnish Environment Institute, Latokartanonkaari 11, 00790 Helsinki, Finland
[b] Department of Resource Economics & Environmental Sociology, University of Alberta, Edmonton, AB T6G 2H1, Canada

ABSTRACT

Policy processes to open digital forest data and information are driven by expectations of increased effectiveness and efficiency of forest management, greater transparency of forest decision making, development of new innovations, and transition to bioeconomy. We investigate how interlinkages between the governance of forest information and governance of forests are being reshaped in the formation of new institutions for open data and information in the case of Finland, where the Forest Information Act was revised in 2016–2018. A qualitative content analysis of public statements related to the legal reform was conducted to understand the perceived benefits and risks associated with more open forest data and information by different actors, and how those perceptions shape their views on appropriate governance of forest information. The analysis reveals conflicts between right to information and right to privacy; concerns about data format, access and usability; as well as the interests of actors with entrenched positions in Finnish forest governance. The debate on opening forest information reflects tensions related to a transition towards greater openness and diversity of values in the forest sector. We envision further research on the relationship between the governance of forests and governance of forest information to support informed decision making during the current open data boom.

1. Introduction

Access to information for forest management has been revolutionized by the development of remote-sensing methodologies for collecting forest data, especially through the emergence of airborne laser scanning applications. In many developed countries with temperate and boreal forests, these technologies have added to the wealth of systematized data available through national forest information systems. Until recently, these data and information have typically only been accessible to researchers, government officials and private sector experts. An urgent question now is how to make data and information more available to a wider group of potential users.

Open data processes are driven by multiple expectations of the benefits of increased access to information and knowledge. This includes the development of new innovations, new business opportunities and services, enhanced democratization and citizen empowerment, and more efficient governance, education and research (Huijboom and van den Broek, 2011). More open sharing of forest information may be considered central for the democratization of forest governance and empowerment of forest users to exercise their rights and advance multiple goals in forest management. The results of multi-stakeholder collaboration and social learning in forest management depend upon the ways in which information is controlled, accessed, and moved between parties (Lei et al., 2015). It may also be seen as a key element in the transition from a fossil fuel-dependent development path to one taking advantage of bio-based resources i.e. 'bioeconomy' (Birch et al., 2010; Staffas et al., 2013) in a sustainable and acceptable way, requiring new innovations to take off (Van Lancker et al., 2016).

Current discussions on the opening of natural resource data often seem to be dominated by a focus on the technical aspects. Yet from the perspectives of environmental governance and institutional analysis, data opening prompts a whole host of new questions. For instance: who is included and whose voices are considered in debates about the governance of forest data and information? How does the distribution of power shape new institutions for data access, management and quality control (cf. Bennett et al., 2018; Prainsack, 2019)? What kind of institutional diversity should be considered in opening forest data and information in different contexts, to support legitimate and effective governance of data and of forests? Though we frequently talk about "opening" data and information, it is rarely the case of fully open access, but in practice some rules – institutions – still condition how the informational resources are shared (cf. Prainsack, 2019).

The policy and scholarly attention to the two extremes of fully open data versus fully closed, intellectual property-protected data overlooks the many governance opportunities that lie in between and what may be learned from e.g. the governance of knowledge commons (Benkler,

* Corresponding author.
E-mail addresses: salla.rantala@ymparisto.fi (S. Rantala), brent.swallow@ualberta.ca (B. Swallow), riikka.paloniemi@ymparisto.fi (R. Paloniemi), emrait@utu.fi (E. Raitanen).
[1] Present address: Suomen kasarmi 1 A 26, 13,100 Hämeenlinna, Finland.

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2004; Boyle, 2008; Frischmann et al., 2014; Hess and Ostrom, 2007). In practice, the governance of opening forest information takes place over different scales, involving a variety of actors and formal as well as informal institutional structures (cf. Frischmann et al., 2014). Complex combinations of contracts, social norms, informal rules and routines, in addition to property rules, frequently comprise the frameworks through which data and information are de facto shared. Which combinations work best to facilitate socially accepted and economically and ecologically sustainable forest management in each context remains a vast, largely unexplored area of enquiry.

The need to revise existing legal frameworks to better match with the policy objectives and technological readiness of opening forest data and information has stirred up heated public debates in countries like Finland, where high-quality forest information is produced at the national level and there is political pressure to better utilize that information for bioeconomy development. These debates bring to fore the multiple interests and motives related to forest data and information which, we argue, are strongly linked to interests in the governance of the actual forest resources. In particular, perceptions of values that may be appropriated from the open data and perceptions of the potential costs or risks related to opening data are likely to shape different actors' views on the ‘appropriate openness’ and governance of forest information. Whose views count in the data policy and legal reforms is shaped by the institutional structures that condition inclusion and participation in the related processes, as well as the power relationships in forest governance that may have been perpetuated over time.

Recently, the application of conceptual and analytical tools developed in natural resource governance scholarship has been explored in the context of emerging knowledge governance regimes (cf. Frischmann et al., 2014; Hess and Ostrom, 2007; Prainsack, 2019). We envision a further step: because natural resource data governance affects the governance of tangible natural resources like forests, and vice versa, we need to study how natural resource and knowledge governance are not merely analogous but inextricably linked. Those interlinks are illustrated in Fig. 1. There are feedback and feed-forward relations between forest governance and forest information governance, driven by the interests of and power relations among actors and by the public and private values that are generated by the forests and forest information (Fig. 1). Increased openness and digital solutions may function as drivers in reshaping the relationships and interlinkages.

In this paper, we thus explore the interlinkages between governance of forests and governance of forest information amidst a policy trend towards more open digital data. We begin by reviewing certain key concepts related to analyzing the benefits, risks and governance of open forest data, and then draw from the case of Finland, where the law on forest information was revised in 2016–2018 as a response to EU regulatory requirements and as part of a national policy agenda to further improve digital forest services and advance bioeconomy development. Through a qualitative content analysis of public and media statements made by various organizations in response to the law reform proposals in 2016 and in 2017, we investigate the perceived benefits and risks associated with more open forest data and information by different actors, and how those shape the actors' views on appropriate governance of forest information. Particular sticking points arise from the Finnish case, related to a conflict between right to information and right to privacy; data format, access and usability; as well as the interests and motives of the involved actors that have their roots in the current and past dynamics of Finnish forest governance. We conclude by proposing areas for further research concerning the relationship between the governance of forests and governance of forest information, to support academic and policy debates during the current open data boom.

2. Key concepts

2.1. Forest/information governance

The study of forest governance is concerned with private and public actors, institutions, and their interplay in light of the effects on forests (Giessen and Buttoud, 2014). Institutions are defined as rules and conventions that facilitate coordination among people regarding their behavior (Bromley, 1989). In the context of forest governance, institutions may include “all formal and informal, public and private regulatory structures, i.e. institutions consisting of rules, norms, principles and decision procedures concerning forests, their utilization and their conservation” (Giessen and Buttoud, 2014, p. 1). We follow a similar conceptualization of governance and institutions in the context of governing informational resources (cf. Frischmann et al., 2014; Hess and Ostrom, 2007).

The shift in focus from ‘governments’ to ‘governance’ essentially acknowledges a broad set of actors influencing forests and relevant policies (Arts, 2014). The actors, ranging from forest custodians such as private forest owners, local communities and state forestry agencies to forest-based industries, governments, NGOs and private citizens, are expected to be at least partially overlapping in forest governance and in forest information governance (Fig. 1) because of intertwined motives related to the values of forests and of data (see 2.2. below). The processes of institution formation for open forest information may also attract attention by other actors who are generally interested in open data questions. The interplay of actors and institutions may be examined from the perspective of new institution formation shaped by actors, their discourses and power relations, and from the perspective of institutional structures that condition participation opportunities and influence of actors (Arts and Buizer, 2009; Bennett et al., 2018; den Besten et al., 2013; Rantala and Di Gregorio, 2014).

In this paper, when examining the inextricable links between governance of forests and governance of forest information, we focus on concepts related to values associated with benefit flows, and the basic rights and capabilities shaping them. Examining the links between institutions for resources and information, we need to understand the nature of public and private benefits generated by data and information, the technologies and mechanisms that society uses to generate and manage information, and the social institutions that enforce openness and protect data privacy.

2.2. Values related to forest data and information

The motives and interests of actors related to open data and information depend upon the private and public values that can be generated and appropriated with that data and information. Private values are benefits that are exclusive to and fully appropriated by particular individuals or firms. Economic theory characterizes such values as excludible and rivalrous (Tietenberg and Lewis, 2015, p. 30). Private values of open data could include, for instance, more rapid development of a new innovation or a method for more cost-efficient targeting of customers. Public values are benefits that are not excludable and not rivalrous. Benington (2011) provides a cogent discussion of four dimensions of public value that are relevant for examining the benefits of open data and information. They include: 1) economic value – generating new economic activity, enterprise and employment; 2) social and cultural value – contributing to social cohesion and cultural identity, individual and collective well-being; 3) political value – stimulating democratic dialogue and active public participation; and 4) ecological value – promoting sustainable development and reducing ‘public bads’ like waste and environmental degradation. Many types of data and information may be combinations of public and private goods, generating some benefits that are excludable and rivalrous, and other benefits that are non-excludable and non-rivalrous.

Environmental and forest data can generate private values and all

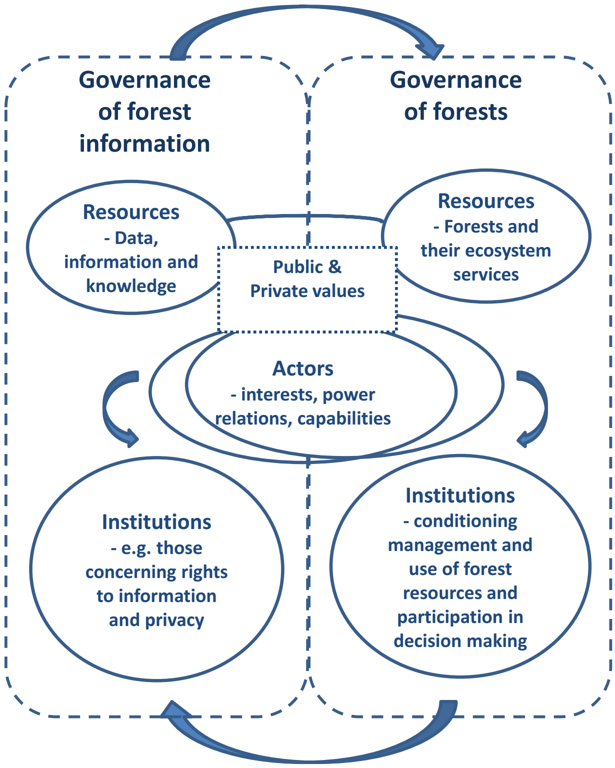

Fig. 1. Interlinkages between forest governance and forest information governance: key concepts.

four dimensions of public values. Open forest data can inform management decisions, such as harvest time, species selection, or designation of critical habitat for conservation of endangered species. Different combinations of economic, social/cultural, political and ecological values can be the result. Open data may also reveal negative impacts from forest cover loss (e.g. Islam et al., 2019), environmental injustices, or even environmental crimes. Actions that follow from the release of such information may in turn lead to more effective management by responsible public agencies for the public benefit in terms of social and ecological values. At the same time, those held responsible for environmental injustices or crimes may be negatively affected in various ways, including fines, loss of operating permits, or loss of credibility and social license with investors or clients. As such, the public and private values of open data and information are often intertwined so that greater openness may result in both winners and losers, with consequences for multiple economic, social, cultural, political and ecological values. We therefore expect that different actors' perceptions of the potential benefits that may be gained from opening data and information, on the one hand, and of the risks of possible losses, on the other hand, greatly affect their views on the 'appropriate openness' of data and information. The extent to which each actor engages in lobbying for their preferred policy option(s) probably depends on the severity of the perceived risks or the importance and likelihood of the potential benefits.

We also expect public actors such as government agencies, and also e.g. non-governmental organizations, to be more concerned with the public values of open data and information, and private actors, e.g. companies, to be more preoccupied with the private values of open data (though the private sector, too, may participate in creating public value and is frequently expected to do so). In many ways, values and value appropriation are both essential and problematic aspects of open data. Private entities can be required to generate and avail data that produces public values, but will have most interest in generating high quality data that has some potential to also generate private values that those entities can appropriate. Data that do not generate private value may be more appropriately provided by non-government or governmental organizations. Economic theory tells us that publicly funded agencies should focus on the provision of valuable public goods and services – public values that are non-excludable and non-rival – that would not otherwise be provided to the market by private suppliers (Pannell and Roberts, 2015). An important aspect is the transaction costs of producing, processing, delivering and maintaining open data that can be justified only if the data is generating discernable public or private value. Governments are always pressed to cut public spending and are in a better position to provide, for instance, open forest data if those data generate both private and public values.

2.3. Capabilities to access open data and information

The varying capabilities of users to access open data draw attention to the formats in which informational resources are shared. The distinctions between data, information and knowledge are important here. Data are raw bits of information, for instance stock volume or height of trees. Information is organized data in context, e.g. in the form of a map. Knowledge comprises the assimilation of the information and understanding of how to use it (Hess and Ostrom, 2007 p.8; citing Machlup, 1983). In other words, the amount of processing and interpretation increases along the gradient from data to knowledge. National forest information systems typically include data and information, while they may be packaged in different ways to contribute to their users' knowledge. The opening of data places the greatest requirements on the users' capabilities, including material resources such as compatible software or financial means to access the software needed for accessing the data, in addition to knowing how to use it and interpret the results. Commercial actors are likely to have an advantage in this over individual citizens, suggesting that most of the private value from open forest data is appropriated by companies and businesses. At the same time, firms also need to constantly upgrade their resources and capabilities to sustain their competitive advantage while responding to the changing environment (Näyhä, 2019). On the other hand, the more the data is interpreted and pre-packaged prior to opening, the more it is likely to influence the users' understanding of the data in certain ways and the application potential of the data.

The public agencies in charge of delivering open data may have an important role in moderating power differentials arising from the different capabilities of actors to use open data (cf. Prainsack, 2019) by striving to make the data and information easily accessible to diverse audiences. However, it is likely to be costly. In addition, the data that public agencies have collected for their own uses may include e.g. estimation methods that must be mastered in order to use the data correctly. This amplifies the importance of high quality meta-data. Any inaccuracies or imprecisions in the data are typically seen as the responsibility of the organization delivering i.e. opening the data, regardless of the initial data source, increasing the quality control burden on public agencies involved in opening data and information.

2.4. Basic rights and legal principles

Relevant basic rights that concern the opening of forest data and information include the right to privacy and the right to information, both seen as vital components of democracy (Rissman et al., 2017). Other relevant rights may be defined in national constitutions and legislation. For instance, the Constitution of Finland secures the right of individuals to participate and influence decision making on their environment, seen as a right equal to e.g. protection of property (Korpisaari, 2018). The right to have environmental information may be considered a fundamental premise in implementing environmental protection – especially participation rights that enable democratic citizenship (Kumpula, 2006).

In many instances, the right to information and the right to privacy conflict (Korpisaari, 2018; Rissman et al., 2017). Because public data are often aggregations of data about individuals, opening such data must account for legislation designed to protect the privacy of individuals. Spatial data on forests may contain information linking forest sites to the individuals owning forest property, the opening of which is limited by personal data regulation. What is considered personal information varies – for instance, the European Union General Data Protection Regulation (2016/679) that came into force in May 2018 defines personal data very broadly as all kinds of data by which an individual may be identified through reasonable effort, including e.g. computer Internet Protocol (IP) addresses and physiological, economic or cultural attributes, in addition to more traditional identifier information such as name, address or social security number. This means that the kind of spatial information falling under personal data regulation may need to be defined on a case-by-case basis (Korpisaari, 2018).

Tackling the conflict between the right to information and right to privacy in effective and legitimate ways is important for improving the governance of private forests for both private and public benefit. For instance, there is increasing interest in using open digital data to advance conservation on private forest lands (Rissman et al., 2017).

3. Case: opening forest information in Finland

3.1. Background: The context for opening forest data and information

About 60% of forestland in Finland, approximately 10.5 million hectares, is privately owned by families. The average size of a family forest holding is fairly small, about 30 ha. The number of forest owners is greater than the number of holdings, since forests are typically owned by spouses together. It is estimated that there are around 632,000 persons owning at least two hectares of forest in Finland (11% of the population of the country). State-owned forests comprise 25% of the total forest area, company-owned forests 9% and the remaining 6% is owned by municipalities and community groups such as parishes (Natural Resources Institute Finland, 2015).

The Finnish national forest information system is one of the most comprehensive in the world. Systematic data have been collected for almost a hundred years, and current remote-sensing based methodologies have further increased the quantity and quality of the available data.[2]

Forest information concerning private lands is produced, maintained and managed by the Finnish Forest Centre under the Ministry of Agriculture and Forestry. The forest information system includes data on forest resources down to the level of individual stands, based on remote sensing and field sampling, as well as information on forest management and use, land use and land ownership. The forest information system covered 82% of the private forest area in 2017.[3]

Since 2012, the Finnish Forest Centre has been developing a web portal, Metsään.fi[4] to facilitate access to private forest information by forest owners as well as service providers. Metsään.fi includes informational resources that can be described as data (e.g. stock volume and age of forest stands), information (maps, habitat information) and knowledge (recommendations on forest management and use by the Forest Centre). By registering as users, forest owners may access these informational resources concerning their own forest holdings online. At Metsään.fi, forest owners can also agree to share their forest information with service providers, such as forestry companies, forest management associations and timber buyers, with a simple click of the mouse. Data may then be directly transferred to the service providers that have signed up for Metsään.fi using a digital communications interface and used for e.g. targeted marketing purposes.

Initially, forest owners had to pay a yearly fee (60€) to use Metsään.fi, but since 2015 they have been able to use it for free. In 2017, Metsään.fi also became free of charge to registered service providers. Along with a gradual expansion of the informational content and extensions of the service, free access is expected to increase user numbers, accelerate information sharing on private forests and boost the efficiency of forest management[3,5]

3.2. Overview of the law reform process

The development of digital forest services in Finland was one of the "spearhead projects" of Prime Minister Sipilä's government within the bioeconomy theme in 2015–2019 (Government of Finland, 2015). The development of Metsään.fi is also a key component of implementing the National Forest Strategy 2025 (Ministry of Agriculture and Forestry of Finland, 2015, p. 30) as a strategic action to promote the quality and availability of forest information. These forest sector developments are part of a broader policy reform to promote open data and digital solutions in Finland, including a cross-sectoral open data program in 2013–2015 (Ministry of Finance of Finland, 2014) and a policy goal to make open science – open publishing, sharing of data and research infrastructure – as the norm in research (Ministry of Education and Culture, Finland, 2014).

The need to reform the legal framework regulating access to private forest data and information was acknowledged in the government's plans in its 2015 strategy document (Government of Finland, 2015). The data produced by the Forest Centre – with public funding – was considered to be personal data of the forest owners, regulated by the Forest Information Act of 2011, subject to the Personal Data Act (1999). Agreement by the forest owner was a prerequisite for third-party access to information concerning any particular forest property. One of the main functions of Metsään.fi initially was to facilitate the pooling of consent to data access by forest owners.

The legal reform process got a boost in September 2015, when Finland received a Written Warning from the European Commission (EC) concerning the lack of openness of forest data. According to the EC view, the data produced by the Finnish Forest Centre was considered public environmental information which should be openly available in compliance with the Directive 2003/4/EC on public access to environmental information.[6] In response, the Government of Finland launched a process to reform the Forest Information Act, led by the Ministry of Agriculture and Forestry.

In the autumn 2016, the Ministry of Agriculture and Forestry invited key forest governance actors in Finland to comment on the first draft bill. The draft sought to comply with the EC requirement to open private forest information, which was strongly opposed by the Central Union of Agricultural Producers and Forest Owners (MTK), the main forest owner interest group in Finland. The first draft proposal would have given forest owners the right to request data on their forest holdings to be removed from public data banks and subsequently information services such as Metsään.fi. This was a compromise solution negotiated during the proposal drafting phase by the major forestry players in Finland; the government, MTK, and the forest industry that had lobbied for full, unlimited access to all forest data and information. The round of comments, however, revealed that the proposal was judicially problematic and even unconstitutional. Following recommendations by the Ministry of Justice and the Attorney General, the proposal was scrapped and the process was started all over, much to the frustration of some of the players involved.

[2] https://www.metsakeskus.fi/more-10-million-hectares-mapped-details-privately-owned-finnish-forests-best-world

[3] Maaseudun Tulevaisuus (MT) newspaper, 28th February 2017 (in Finnish).

[4] "To the forest"; https://www.metsaan.fi/en/briefly-english

[5] https://www.metsakeskus.fi/content/metsaanfi-muuttuu-maksuttomaksi-toimijoille-132017 (in Finnish).

[6] https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:32003L0004

The second proposal for the new Forest Information Act was introduced in late 2017. In this proposal, grid data (16 × 16 m) on forest resources, including, for instance, main tree species and height of trees, was to be open access to anyone without the need to register as a user or to solicit consent by forest owners. Also environmental information on critical forest habitats as well as sites of planned forest management or logging operations, based on mandatory notifications to the Forest Centre, was to become open access. Access to information that may be connected to the personal identity of a forest owner such as name, contact information or social security number would require a specified request to the Forest Centre. However, other reasons would be considered valid in granting access, such as a contract or a membership, in addition to the forest owner's direct consent given in Metsään.fi. Public organizations could also directly access the information based on relevant sectoral laws without case-specific consent. An interesting new feature was that the law enabled inclusion of data and information produced by other actors besides the Forest Centre in the database, an example of crowdsourcing. This proposal largely satisfied most actors and was passed on to the Parliament where a revised version of the law was approved.

The new law came into force on March 1st, 2018. Immediately, a new feature in Metsään.fi allowed anyone to browse and download open grid data concerning private forests without registering as a user. The registration-based more detailed services described above are still available for forest owners and service providers.

3.3. Material and methods

A mandatory element of the Finnish legislative drafting process is soliciting stakeholder comments on draft government bills. The draft bill along with a request for comments is sent to known key stakeholder groups and the request is also published so that other interested parties have opportunities to comment. A summary of the received statements is published and included as background material in the government proposal.[7] We used the statements submitted through this procedure in 2016 and 2017 (see Section 3.2 above) as material to explore the benefits, risks and institutional preferences associated with the forest information legal reform by different actors. In addition to the actual draft bills, the material included six statements submitted by different organizations concerning the draft of 2016 and eighteen statements submitted regarding the proposal of 2017, as well as two summaries of the received statements and six press releases related to the process issued by the Ministry of Agriculture and Forestry. The statements from 2017 were readily available on the parliament website and could all be included, but the original statements related to the draft proposal of 2016 could only be recovered to the extent that they were available through the respective organizations that had made the statements.

These materials were complemented by searching the electronic archives of the two most widely circulated national-level newspapers for articles related to the law reform process, resulting in the inclusion of 48 items (editorials, columns, news items, letters to the editor) published in the newspaper *Maaseudun Tulevaisuus* (MT; "The Rural Future" in English) in 2016–2018 and three editorials from Finland's most-read newspaper, *Helsingin Sanomat*. The interest of MT is explained by its strong focus on rural affairs, agriculture and forestry, though it also covers news broadly and is the second most-read newspaper in Finland (MediaAuditFinland, 2017). Furthermore, according to its website, MT is formally linked to the Central Union of Agricultural Producers and Forest Owners (MTK), one of the key actors in the forest information debate. Also in the case of the media material, the focus of the analysis was strictly on the views expressed by stakeholders in interviews, public events, or in the content produced by these stakeholders themselves (columns, letters to the editor), rather than media representations on opening forest information.

A qualitative content analysis was carried out using the Nvivo software (QSR International Pty Ltd. Version 12, 2018; Bazeley and Jackson, 2013). The material was coded in three phases; first, text excerpts were extracted from the material following a broad categorization concerning the benefits, risks, views on governance of open forest data and information, and the law reform process. Second, inductive coding was applied to material within the broad first-tier categories. An inductive approach was chosen to capture the diversity of views related to the potential benefits, risks and institutional preferences regarding the opening of forest data and information. The codes were a mix of more interpretative labels intended to represent a common meaning for similar arguments and labels directly corresponding to specific, recurring expressions found in the data. The codes were then further grouped into themes that were connected back to the key concepts outlined in Section 2 of this paper. In the following section on the results, the views on governance and institutional preferences are presented focusing on the themes that emerged as most salient and divisive through the inductive coding.

4. Results

Though various organizations took the opportunity to comment on the law reform proposals in 2016 and 2017, most supported the intent of the proposals. It is clear from the media material that the actual debates and negotiations had happened outside of the formal commentary phases, among the actors with the most clearly contrasting views on the benefits, risks and governance of opening forest data and information: the government, represented by the Ministry of Agriculture and Forestry and the Forest Centre; MTK; and the forest industry, represented most vocally by the Finnish Forest Industries Federation. They are the usual key actors in Finnish forest governance (Kotilainen and Rytteri, 2011; Kröger and Raitio, 2017; Paloniemi and Varho, 2009; Rantala and Primmer, 2003). Other frequently active forest governance players, such as nature conservation NGOs, also voiced their views but to a lesser extent. This might reflect their general satisfaction with the seemingly inevitable governance change towards greater openness, as could be discerned from their statements.

It is also noteworthy that despite the formally open call for comments, the organizations that commented on the proposals were largely forest sector organizations, conservation NGOs and public agencies (ministries). Only a few actors that could be considered to have a specific focus on information governance submitted their comments (Open Knowledge Finland, the Data Protection Ombudsman's office, and a private company specializing in spatial data solutions). This resonates with our argument regarding the entanglement of forest resource and forest information governance. The motives of actors to get involved in the institution formation processes for forest data and information are likely to be closely related to their interests concerning the actual forest resources.

4.1. Benefits

Arguments related to the public economic values of open forest data – adding value to the forest data produced as a public service – were salient in the statements of the key government actors, the Ministry of Agriculture and Forestry (in charge of preparing and facilitating the law reform process) and the Forest Centre (responsible for producing and managing the data and the related technological structures). This echoed the formal government strategies. Opening private forest data was argued to support digital and wood-based bioeconomy, enabling new innovations, fostering business opportunities and eventually creating a competitive advantage for the overall Finnish forestry sector. Greater foreign investment in the forestry sector may result. It was also proposed that more open data would stimulate more active forest planning and management, through better and more targeted services

[7] http://lainvalmistelu.finlex.fi/en/3-lausuntomenettely/#esittely

to forest owners. It was especially hoped that these more targeted services would better mobilize "dormant" or absentee forest owners who increasingly live in cities and do not rely on forests for their income. Improved cost-efficiency of the forest administration system was another salient topic, particularly in relation to the duties of the Forest Centre.

Ecological values were also emphasized in the government actors' statements. Potential of open data was expressed in relation to forest law enforcement and increasing the sustainability of forest management, for instance, in the monitoring of compliance with the legal requirement to replant logged areas and of the development of the nature values of logged and restored areas. The Ministry of the Environment highlighted how open forest data could enable taking nature values into account already in the planning phase of forest operations, enhancing the sustainability of extractive forest industries. Furthermore, new livelihood opportunities could be created based on non-extractive forest uses, such as nature-based tourism.

The Forest Centre also drew attention to the potential of open forest data to improve the transparency of forest decision making and administration, and to increase public participation in debates about forests and their management. Together with the improved state forestry services, this could improve the legitimacy of forest management overall. The same political values related to improving the transparency and legitimacy of forest decision making were also salient in the statements of conservation NGOs, who emphasized the need to meet people's right to information about their environment, in line with the national legislation and the EU directive.

Unsurprisingly, economic values were central also in the arguments of forest industry actors, represented most actively in the debates about opening private forest information by the Finnish Forest Industries Federation and backed by the Finnish Sawmills Association and the Trade Association of Finnish Forestry and Earth Moving Contractors. They, too, emphasized an outstanding chance for the forest industry, forest owners and the public sector to develop collaboration and new innovations based on open data and digital services, leading to international competitive advantage in developing a diverse and sustainable forest industry. More direct marketing and better services would bring benefits to forest owners and mobilize more wood from forests as a result of more effective forest management, contributing to related policy goals. In a counter-argument to MTK, an industry representative stated that they would have more money to buy wood from forest owners if less was spent on finding it (MT, September 21st, 2016). In other words, open data was seen to have potential to create cost-efficiency benefits also in the private sector. A central argument for the industry actors concerned increasing equity; that all forest service providers and companies, big and small, would have the same access to data and information, which would stir healthy and fair competition in the industry – again to the mutual benefit of forest owners.

4.2. Risks

Risks related to the opening of private forest data and information per se were really only expressed by one actor, the MTK, whose statements were focused on the risks and contained almost no articulation of benefits. Their arguments centered on forest owners' right to privacy concerning forest possessions. Opening private forest information was repeatedly equated with disclosing the bank account balance of a forest owner, presumably because the monetary value of forest holdings could be estimated if open data on forest stock volume and forest ownership were combined with e.g. timber market information from other sources. The issue was elaborated with arguments concerning a weakened position of forest owners in the market compared to industry actors, including fears that large-scale forest owners would be favored over owners of small forest estates.

Whereas other actors saw an opportunity in the opening of private forest data for monitoring the sustainability of forest management, MTK argued that this constituted a risk that outsiders would increasingly interfere with the private decision making of forest owners, especially concerning logging. In their view, information would be accessible also to actors "with a negative view on forest use" (such as Greenpeace) who could draw public attention to plans to log an old-growth forest. MTK blamed the government for "socializing" private property through making private forest information public, and suspected that the other actors, particularly the Forest Centre and the forest industry, were only using EU regulation as a pretext to further their own motives, since the EC was not demanding any other member country to make forest information public.[8] They warned that the current "system" – either referring to the actual forest information system or, more generally, to a carefully negotiated balance between the government, forest industry and the forest owners represented by MTK – was built on mutual trust that would be eroded if the data were opened to all.

Other actors mainly saw risks in the specific approaches presented in the draft bills to limit the full opening of the data in order to satisfy MTK's demands for the protection of privacy of forest owners. These were related especially to the right to erase data included in the 2016 draft, and the pros and cons of limiting open access to grid-level data only, discussed in detail in the following.

4.3. Governance: The sticking points

4.3.1. Nature of data and information on private forests

Strikingly different views on the nature of data and information on private forests were presented in the debates, to justify the approaches preferred by different actors. The forest industry and the NGOs were most strongly in favor of the EC view that private forest information should be treated as public environmental information. According to the Finnish Forest Industries Federation, all publicly funded data should be public, and there was no legal basis to restrict access to any part of private forest information. MTK saw things exactly the opposite: data and information on private forests were private regardless of who were producing them, and forest information was essentially not the type of environmental information concerning citizens' health and well-being that everyone should be able to access according to the EU directive. Contesting the main argument in the government's plans and proposals, MTK argued that fully open data was not required to advance the bioeconomy. According to MTK, private forest information was information on livelihoods, comparable to a business plan, which should only be accessible with the individual forest owner's consent.

During the round of comments in 2017, after the key actors had largely conceded to the latest proposal, the Data Protection Ombudsman's office was the only one to note shortcomings in the preparation of the draft bill, to the extent that they recommended shelving it. They drew attention to the definition of personal information in the draft and the legal space to sideline protection of privacy regulation through sector-specific legislation, such as the Forest Information Act. The Ministry of Agriculture and Forestry countered these views, stating that protection of privacy had been sufficiently addressed in the draft through defining which personal data could not be openly accessed. Moreover, the Ministry of Agriculture and Forestry declared that forest data and information were not within the core area of privacy protection: "information on forests as well as forest management and use is such information which is easily obtainable and does not as such reflect anyone's personal conditions. Because of everyman's rights,[9] anyone can move freely in forests. Therefore anyone

[8] However, the same EU directive underlies the Swedish Forest Agency's decision to publish notifications on planned forestry operations since 2016 (https://www.atl.nu/skog/eu-varnar-finland/, in Swedish).

[9] Everyman's rights refer to principles in Finland and other Nordic countries that secure everyone's access to and enjoyment of natural areas for recreational purposes on public and private land, with few limitations, save for private yards

can make observations about the forest environment and operations therein" (Ministry of Agriculture and Forestry, 2017). While the scope of application of personal data regulation cannot be defined in sector-specific legislation, the interpretation that forest information is not core personal information enables the drafting of sectoral legislation that defines the procedures for handling that information. The Ministry of Justice found no constitutional or other judicial problems with the 2017 proposal.

The differing views show how the interpretation of forest information as a resource was linked to different conceptualizations of private forests as a resource. MTK interpreted private forests primarily as a source of livelihoods, fully within the domain of private decision making and only secondarily relevant for the public values. Government, industry and NGOs focused on the public ecological, economic and social values of private forests and access enabled by everyman's rights. Industry actors also stressed that public funds were used to produce the informational resources, implying a public stake in them, while MTK saw the *object* of the data and information, private forest property, as the most important element for defining appropriate governance of the informational resources.

4.3.2. Right to erase data

In response to MTK's concerns about the protection of privacy and of property, the first draft proposal in 2016 included a clause that would have allowed forest owners to request the data concerning their forest holdings to be erased from the public databanks. MTK lauded this decision to give the last word to the forest owner. Other actors found it problematic, especially from the point of view of data cohesion and the quality of the national forest information system. For instance, the monitoring of the state of forests would be compromised if part of the information was erased and data was no longer comparable to those of the previous years. Environmental NGOs, the Ministry of the Environment and the Forest Centre also saw the proposal as threatening law enforcement, especially due to potential removal of information on critical forest habitats. In addition, the Forest Centre was concerned about a weakened capacity to carry out its duties, including targeted advisory services and communications to forest owners in order to improve the management of private forests.

The data removal possibility was critiqued for not being in line with the spirit of the EU directive or national legislation. The Finnish Association for Nature Conservation argued that the right to erase data was poorly founded on forest owners' conceptions of what were private data and therefore not judicially solid. The Ministry of Justice and the Ministry of Finance pointed out that it meant a departure from the requirement for data integrity of public information systems defined in the Act on the Openness of Government Activities (1999). Finally, the right to erase public data was considered contrary to the principle of openness in the Constitution (12 §) and subsequently discarded.

The data removal debate can be seen as an extension of the different interpretations of the nature of forest information, reflecting actor interests in the governance of the actual forest resources. MTK valued private decision making of forest owners above all, while the other actors were concerned with a potential weakening of the role of private forests in producing public values and especially the authorities' capacity to support and to keep track of it. In that sense, the right to erase data would have essentially worked against some of the expected benefits for forest governance as articulated by the government actors (4.1).

4.3.3. Open data format: Stand vs. grid-level data

Another highly divisive matter in the debates concerned the format in which the data was to be opened: grid- or stand-level data. Grid data is the raw data on 16 × 16 m areas produced through laser scanning and aerial photography which can be further processed to produce information at the level of forest stands. Stands are frequently the unit of forest management and operations, consisting of fairly homogenous trees or trees that share a common set of characteristics, and varying in size. The stand-level data in the information system of Forest Centre is connected to information on the boundaries of forest estates and forest ownership. The opening of stand-level data was opposed by MTK because of the link to personal information and what they saw as the associated risks. According to them, the opening of grid data would fully satisfy the EC requirements for open environmental information and serve the development of new digital services in the bioeconomy. MTK also argued that focusing on the production of grid-level data would ease the work volume on the Forest Centre resources so that data could be updated more frequently. The production of stand-level data could be left to private actors with the interest and resources to do it.

The Forest Centre defended the production and sharing of stand-level information as the common language between forest owners and service providers, as stand is the unit of management that all were used to. They pointed out that few actors could make use of the raw grid data, requiring a system to process and interpret the data, other than large forest companies and the regional forest management associations under MTK. In order to serve all information users equitably, they argued, it was essential that the Forest Centre continue producing stand-level data. Besides, they pointed out, if the delivery of stand-level information was left to the private sector, then what would be the possibilities for forest owners to control it?

The forest industry actors were also strongly in favor of the opening of stand-level information as a key element in supporting and enhancing interaction between forest owners and businesses. They highlighted the costs of further processing grid data which cannot be directly used to market services related to forest use or operations. The Finnish Forest Industries Federation cited the Finnish Competition and Consumer Authority in that opening only grid data would work to the benefit of forest management associations and complicate the entry of new actors, development of new services, and promotion of competition in the field. As such, the format of open data, in addition to its content, was seen to greatly affect conditions for fair competition in the forestry sector. The Trade Association of Finnish Forestry and Earth Moving Contractors also pointed out that large forest-sector companies and forest management associations were best placed to utilize grid data, while small players would need to invest in new ICT to access the same opportunities from open data. Therefore, the continued delivery of stand-level information was important in order to avoid the risk of market concentration among fewer actors.

Drawing attention to the rapid technological development of remote-sensing based forest inventory methods, the Forest Centre suggested that the production and delivery of forest data should not be legally tied to any specific unit of measurement, since the grid might turn out to be an unsuitable unit in the future, as inventory methods become ever more sophisticated. They feared that regulation based on specific methods would almost inevitably lead to problems of legal interpretation and in the worst case, hinder development to enhance the cost-efficiency of data production and management.

Also the tug-of-war over grid and stand-level data appears to coalesce around privacy and private decision-making in forestry. But clearly economic values were at stake as well; the expected cost-efficiency benefits from open data for the private sector could be partially undermined by the processing costs of grid data, favoring the actors already equipped to do so, such as the regional forest management associations. Thus, in addition to the privacy concerns that dominated the public argumentation of MTK, economic interests in forestry were also reflected in the institutional preferences for opening forest information.

In the end, the new law governs the production and sharing of both

(*footnote continued*)
and cultivated areas. Cf. http://www.ymparisto.fi/en-US/Nature/Everymans_rights%2827721%29

grid- and stand-level data, and the Forest Centre continues to update stand-level data based on modeling and information on forest management and operations. However, stand-level data has been opened in a more limited manner and can only be downloaded as packages at regional and municipal level, while grid data can be directly browsed on a server map in addition to being available for download.

4.3.4. *Crowdsourcing*

An issue that also received comments in several statements, though it proved not to divide opinions so much, concerned the new opportunity provided by the law to include data and information produced by anyone in the Forest Centre's database, as long as the quality has been controlled by Forest Centre staff. In general, the inclusion of crowdsourced information was welcomed, and it was seen to have great potential to speed up the update of information on, for example, biodiversity as well as storm-related or other forest damage. However, concerns were also raised regarding the quality control of such information and the associated extra workload for the Forest Centre. It was suggested that the criteria that the Forest Centre uses to screen potential deficiencies or errors in the data should be defined openly and transparently. The Natural Resources Institute Finland highlighted cyber security issues and suggested that the Forest Centre pays specific attention to potential manipulation of information and intentional spread of misinformation in handling crowdsourced data. The Natural Resources Institute recommended that data and information should only be accepted from sources that can be verified and provide appropriate metadata to facilitate the quality control.

5. Discussion

The opening of digital data and information on forests and other natural resources is a policy trend that presents a new area of research for environmental governance scholars. Our empirical findings on the changing governance of digital data and information on private forests in Finland demonstrate the tensions related to a transition towards greater openness and diversity of values in the forest sector.

At first glance, the debate about opening forest data and information in Finland appears to be about a conflict between the right to information and the right to privacy. Both feature extensively in the public arguments of the involved actors, the right to information perhaps more strongly backed by the relevant institutional framework, including the EU directive on public environmental information. Though the push for greater openness is enjoying extensive policy support at the moment, more nuanced demands for a balance between privacy and openness can also be observed,[10] especially as breaches of personal data protection associated with big data are publicized.[11] Undoubtedly, the debate about right to information versus right to privacy will be re-invented repeatedly in the near future. In the EU context, different countries have to determine what the General Data Protection Regulation (2016/679) means for handling spatial data on forests and other natural resources.

The case of Finnish private forest information also illustrates how multiple values related to the data and information by various actors, linked to interests in the governance of the actual natural resources, play an important part in the formation of institutions for opening data. The frequently mentioned benefits of open data, especially related to public values, were salient also in this case. Public values were understood to include more effective and cost-efficient provision of government services; greater collaboration between public, private and civil society organizations; and more effective, transparent and accountable management of forests. For the private sector, reductions in cost and improvements in efficiency emerged as especially important. In the public sector, reduced costs create public economic benefits. A key victory for those in favor of greater openness was the opening of information on critical forest habitats and planned forestry operations. In contrast, the arguments of MTK reflect a perception that more open data will be part and parcel of a new set of limitations of private rights to use forest resources. How future public debates and decision making on forest management are shaped at the local and national level, now that the public may increasingly act as a watchdog of forest use based on the open information, constitutes an intriguing area of future research.

The capabilities aspect was strongly reflected in the debate about grid and stand-level data. The continued provisioning and opening of stand-level forest information was justified as the common language that enables communication between the different actors in the forest sector and moderates equal access to benefits from open data. In contrast, limiting openness to grid data was seen to favor actors that can invest in further processing the data. Thus, we found that capabilities and economic values of forest information were strongly related.

Pulling all these aspects together, deeper connections and change dynamics between the actors and institutions in Finnish forest governance and the governance of forest information may be uncovered. To better understand this, one needs to note two dominant factors that explain forest governance in Finland compared to most other countries: the important role that forests play in the Finnish economy and the ownership structure dominated by a large number of private holdings (Natural Resources Institute Finland, 2015; Rantala and Primmer, 2003). In other words, the *status quo* of Finnish forestry – for at least half a century – has been based on private ownership of forests and livelihoods from extractive forest industries. Some actors, as reflected in their public arguments, prefer retaining that system. Path dependency is well-recognized as a challenge for the traditional Finnish forest industries (Näyhä, 2019). However, the debate about forest data and information reveals how the whole sector is in transition, and the multiple values of forest information reflect the increasing diversity of interests and overlapping values related to forests themselves. These overlapping values include, for instance, economic values, public ecological values that are increasingly important also to private forest owners (Pynnönen et al., 2018) and to the general public (Valkeapää and Karppinen, 2013), potential values related to novel uses of forests, and political values including the openness and transparency of forest decision making. Contemporary understandings of value and value creation are also undergoing changes (Näyhä, 2019). Pitting forests as a domain of private livelihoods against a more dynamic, open and diverse view of the forest sector reflects a struggle over definitions and ultimately, power. Illustrative is the view expressed by MTK that “the whole system” is based on trust between them, the government and the forest industry, which will be eroded if data and information on private forests is opened to all. That is, opening forest information means opening forestry, and holding on to the established power bases will become ever more challenging. Accordingly, it is essential to emphasize fairness to all actors to encourage trust building and collective action in data governance (cf. Hotte et al., 2019).

In the forest sector as well as in other domains, open data and information can be seen to have potential to weaken old hierarchical structures by allowing new direct connections in networks of information and communication. Benington (2011) draws attention to a shift away from a government and market dominated society towards “networked community governance” in which the civil society has an increasingly important role. This change also increases the number of actors, perspectives and goals that need to be taken into account while creating rules for data governance. The debate on opening forest information in Finland demonstrates such a diversifying governing community. In a situation with numerous, partly conflicting goals, public debate on priorities is essential (Lindahl et al., 2017).

[10] cf. https://mydata.org/

[11] e.g. “Facebook formally fined £500,000 for Cambridge Analytica scandal”, Financial Times, 26th October 2018. https://www.ft.com/content/2f4593ee-d832-11e8-ab8e-6be0dcf18713

In the market economy, many more service providers can market their services to forest owners based on open forest information, undermining the role of intermediaries who therefore stand to lose as a result of the opening processes. In Finland, the regional forest associations under MTK have traditionally acted as such gatekeepers between forest owners and the industry; a role that could now erode. In general, open data may reduce information asymmetries that have worked to the advantage of forest owners in wood markets (cf. Arrow, 1984), especially through Artificial Intelligence applications developed to analyze mass data (Marwala and Hurwitz, 2017). An interesting question is whether and how open data supports the development of new service concepts offered to forest owners (Laakkonen et al., 2019) that better match the increasing diversity of forest-related values of forest owners, including non-extractive forest uses (Pynnönen et al., 2018). This is especially relevant in the case of the 'dormant', city-dwelling forest owners who do not rely on forests for their livelihoods. At the same time, underlying the policy change in Finland was the government's goal that open data applications also enable the extractive forest industries to reach this category of forest owners more effectively.

The forest data and information in the analyzed Finnish case concerned remote-sensing based forest resource data and the associated information on forest management and ownership contained in the national forest information system. For new radical innovations in forestry and in bioeconomy, the relevant knowledge base is much broader and more complex. In addition to data and information on the natural resources, nuanced information on market opportunities across the value chains, technological know-how and social knowledge that may form the basis of novel and dynamic innovation networks (Allen and Potts, 2016; Van Lancker et al., 2016) are needed. Subsequently, the institutional diversity for governing the knowledge base in support of a digital transformation is more complex, spanning norms, practices and contracts beyond the formal governance frameworks, as much of the relevant information and knowledge is likely to be tacit or protected. New areas of conflict of interests are likely to emerge and need to be resolved as part of advancing the digital bioeconomy.

The Finnish case provides some interesting insights into the interlinkages between the governance of forests and the emerging governance of open forest data, rooted in the history and the specific characteristics of forest governance in the country. The characteristics of forest governance clearly defined the way that different interest groups defended or promoted different public and private values of open forest data, as well as the way that they aligned in the debate about openness versus privacy. In terms of policy implications, the Finnish case further demonstrates how important it is to pay attention to an inclusive process to define the rules of the game while developing readiness to open public data and information, instead of treating them as an afterthought. Probably nobody expected the issue to stir up such polarized debates, while the technological development for sharing forest data was already well-advanced. Yet in light of our results describing the intertwined interests in forests and forest information, it is clear that the governance aspects should not be overlooked while opening data and information. It should be recognized that the actors with the most stake in the resources will also be most involved and committed to debates about related data and information. Transparency, access and privacy will be front and centre in most debates about open data, and we should expect concerned parties to frame their positions on these three topics by the stakes that they hold. On the other side, the effects of forest information governance on forest resource governance will emerge slowly, and it would be important to track for what purposes the information is used, with what effects, and who might lose as a result.

Our study gives rise to a number of questions for further research. First, it will be relevant to analyze what the dynamics of forest data policy processes will look like in other contexts, with different constellations of forest governance actors, institutions and power relations. For instance, how will processes play out in countries where the formal forest policy goals tend more towards conservation than forest use, or in countries where international development actors are significant players in national forest policy? What about contexts where existing forest data and information are scattered or of low quality, and new transnational actors may come in with superior capabilities and advanced technology combined with commercial interests? Second, it will be elemental to study the effects of new forest data and information policies on the governance of forests in those diverse contexts, and the role of open data and digital solutions as drivers of change in the forest sector. We hope to see a surge of studies exploring this emerging area of research, improving understanding of the complex linkages between the governance of data and information and the governance of forests, and how they are being reshaped and transformed by the openness trend.

Declaration of competing interest

None.

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